BIOLOGY | BENCHMARK #1

Open Response: Population Dynamics

Massachusetts Biology Standard HS-LS2-1: Analyze data sets to support explanations that biotic and abiotic factors affect ecosystem carrying capacity.

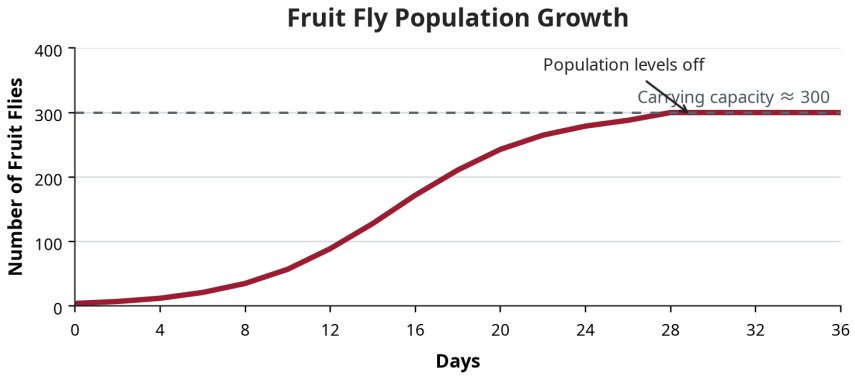
TOTAL SCORE

____ / 10

Name: _____	**Period:** _____	**Date:** _____

Directions: Use the population graph and food web to answer all questions. Write in complete sentences when an explanation is requested. Use evidence from the visuals and your knowledge of population relationships.

PART A | Read the population graph The graph models a fruit fly population in an ecosystem with predators, including frogs, spiders, and mites.



1. Identify the population represented by the graph. [1 point]

The graph represents the _____ population.

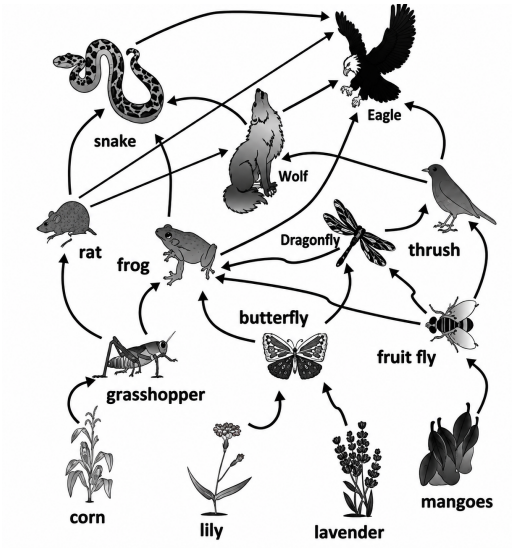
2. Describe how the fruit fly population changes from Day 0 to Day 20. [1 point]

Use a trend word such as increases, decreases, levels off, or stays about the same.

3. What is the carrying capacity, and on what day does the population first reach it? [2 points]

Carrying capacity: _____ fruit flies First reaches it on Day: _____

PART B | Food web and constructed response Arrows in the food web show energy moving from the food organism to the consumer.



4. Use the population graph and food web to explain one factor that could cause the fruit fly population to level off near carrying capacity. [2 points]

You may use predators, resources, space, disease, or competition in your explanation.

5. Suppose the number of frogs increases. Use the food web and population graph to predict how the fruit fly population may change. Write a Claim–Evidence–Reasoning response. [4 points]

Claim • Evidence • Reasoning Make a prediction, cite evidence from both visuals, and explain how the evidence supports your prediction.